

FBISE Class 9th Chemistry Solution 1st Annual Exam, 2026

Code 10051

Sec A MCQs

1. What does chemistry mainly study? B — Matter and its interactions

Matter is anything that has mass and occupies space. Chemistry studies what matter is made of, its properties, and how substances change or react with one another.

- **A, living organisms:** These are mainly studied in biology, though biochemistry studies their chemical processes.
- **C, planets and stars:** These are mainly studied in astronomy.
- **D, historical events:** These are studied in history.

2. Which sample contains Avogadro's number of molecules? C — 44 g of CO₂

Avogadro's number, approximately 6.022×10^{23} , is the number of particles in **one mole**. CO₂ has a molar mass of $12 + 2(16) = 44$ g/mol. Therefore, **44 g of CO₂ is one mole** and contains Avogadro's number of CO₂ molecules.

- **A, 0.1 g of H₂:** One mole of H₂ is 2 g, so this is less than one mole.
- **B, 14 g of N₂:** One mole of N₂ is 28 g, so this is half a mole.
- **D, 10 g of H₂O:** One mole of H₂O is 18 g, so this is less than one mole.

3. What is the sum of oxidation numbers in SO₄²⁻? A — -2

The oxidation numbers of all atoms in an ion must add up to the **charge on that ion**. Sulfate has a charge of 2 -, so the sum is -2.

You can check it: oxygen is usually -2, giving $4(-2) = -8$. Sulfur is +6, and $+6 - 8 = -2$. Options **B (-1), C (0), and D (+2)** do not match the sulfate ion's charge.

4. Why does ammonium nitrate make the container feel cold? C — Endothermic

Dissolving ammonium nitrate **absorbs heat from the water and surroundings**. As the surroundings lose heat, the container feels cold. A process that absorbs heat is *endothermic*.

- **A, neutralization:** This refers to an acid reacting with a base.
- **B, exothermic:** An exothermic process releases heat to its surroundings.
- **D, combustion:** This is a burning reaction.

5. Why might a reversible reaction not go to completion? C — Backward reaction occurs simultaneously

In a reversible reaction, products can react to form the original reactants again. As products form, this **backward reaction** can occur alongside the forward reaction. At equilibrium, both continue at equal rates, so the reaction need not use up all the reactants.

- **A, products never form:** Products do form; some can change back into reactants.
- **B, forward reaction is faster than backward:** That alone does not explain why the reaction stops short of completion. At equilibrium, their rates are equal.
- **D, energy is not involved:** Energy changes can occur in reversible reactions.

6. What forms when an acid reacts with a metal carbonate? A — Salt + CO₂ + H₂O

The general reaction is:



For example:



- **B:** Hydrogen and ammonia are not the products of this reaction.
- **C:** The gas produced is **CO₂**, not nitrogen.
- **D:** Salt forms, but so do **CO₂ and water**.

7. Charge on a Group I-A ion — B: +1

Group I-A atoms, such as sodium (Na), have **one electron in their outermost shell**. They usually lose that electron, leaving a positive ion: $\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$.

- **A (-1):** Would mean gaining one electron.
- **C (+2):** Would require losing two electrons; Group I-A atoms normally lose one.
- **D (-2):** Would mean gaining two electrons.

8. Trend down Group I-A — D: Density increases

Density **generally** increases down Group I-A as the atoms become heavier.

- **A, all become gases:** Group I-A metals are solids at room temperature.
- **B, melting point increases:** Their melting points generally **decrease** down the group.
- **C, reactivity decreases:** Their reactivity generally **increases** because the outer electron becomes easier to remove.

Exam note: “Density increases” describes the *general* trend, not an exceptionless rule; potassium is less dense than sodium.

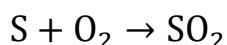
9. Nitrogen in clean, dry air — C: 78%

About **78% of clean, dry air is nitrogen gas (N₂)**. About 21% is oxygen.

- **A (10%) and B (21%):** Neither is nitrogen’s approximate share; 21% is approximately oxygen’s share.
- **D (90%):** Too high.

10. Source of sulfur dioxide — B: Burning sulfur-containing fossil fuels

When sulfur present in coal or petroleum fuels burns, it combines with oxygen to form sulfur dioxide:



- **A, catalytic converters:** They are used to reduce certain harmful vehicle emissions; they are not the source described here.
- **C, melting glacier ice:** Melting ice does not produce SO_2 .
- **D, rusting iron:** Rusting forms iron oxides, not sulfur dioxide.

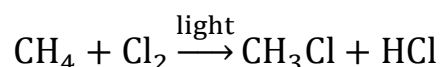
11. More than one structural formula — D: C_4H_8

A *structural formula* shows how atoms are connected. The same molecular formula, C_4H_8 , can have different connections. For example, **but-1-ene** has its double bond at the end of the chain, while **but-2-ene** has it in the middle.

- **A, CH_4 :** One carbon atom gives no alternative carbon arrangement.
- **B, C_2H_6 :** Its two carbon atoms have only one arrangement here.
- **C, C_3H_8 :** Propane has only one carbon-chain arrangement.

12. Typical reaction of an alkane — D: Substitution reaction

Alkanes contain only **single bonds** between carbon atoms. A common reaction replaces one of their hydrogen atoms with another atom, such as chlorine. This is *substitution*:

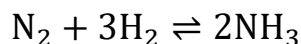


- **A, addition:** Typical of compounds with carbon–carbon double or triple bonds, such as alkenes.
- **B, neutralization:** An acid reacts with a base; that is not an alkane's typical reaction.
- **C, dehydration:** Removes water from a substance; it is not the typical reaction of an alkane.

Sec B Short Questions

(i) Branches of chemistry

a. Large-scale production of NH_3 by the Haber process — Industrial chemistry. Industrial chemistry deals with producing chemicals on a large scale. The Haber process manufactures ammonia from nitrogen and hydrogen:



b. Breakdown of glucose to provide energy in the body — Biochemistry.

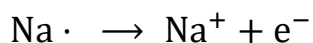
Biochemistry studies chemical processes in living organisms. During cellular respiration, cells break down glucose to release usable energy.

c. CFCs in refrigerators linked to depletion of the ozone layer — Environmental chemistry. Environmental chemistry studies chemicals in the environment and their effects on air, water, and soil. CFCs can release chlorine-containing species in the upper atmosphere that damage ozone.

OR: Show the ion conversions with electron-dot structures

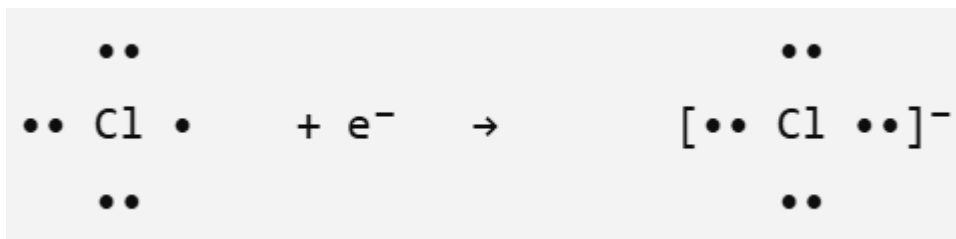
A dot represents a **valence electron**. Sodium has one; chlorine has seven.

a. Sodium loses one electron:



The resulting Na^+ has **one fewer electron than protons**, so its charge is +1.

b. Chlorine gains one electron:



There are **seven dots** around Cl on the left and **eight dots** around Cl^- on the right. The extra electron completes chlorine's outer shell and gives it a -1 charge.

(ii) Density and compressibility of solids and gases

Solids: Their particles are packed **very close together**, with very little empty space. This usually gives solids high density. There is little room to push the particles closer, so solids have **very low compressibility**.

Gases: Their particles are **far apart**, with much empty space between them. A given volume therefore usually contains relatively little mass, giving gases low density. Pressure can push the particles closer together, so gases are **highly compressible**.

Solid	Gas
Particles close together	Particles far apart
Little empty space	Much empty space
Very difficult to compress	Easy to compress

OR: Moles in 100 g of aluminum foil

Use:

$$n = \frac{m}{M}$$

where $m = 100$ g and the molar mass of aluminium is $M = 27 \text{ g mol}^{-1}$.

$$n = \frac{100}{27} = 3.70 \text{ mol}$$

Answer: The foil contains approximately 3.70 moles of aluminium atoms.

(iii) Average relative atomic mass of X

The average accounts for how common each isotope is:

$$A_r(X) = (64 \times 0.30) + (66 \times 0.70)$$

$$A_r(X) = 19.2 + 46.2 = \boxed{65.4}$$

Answer: The average relative atomic mass of X is 65.4. It has no unit because relative atomic mass is a ratio.

OR: Determine the number of Protons and neutrons in the isotopes

Neutrons = mass number – atomic number

Isotope	Mass number	Protons	Neutrons
Sodium-23	23	11	$23 - 11 = 12$
Chlorine-37	37	17	$37 - 17 = 20$
Carbon-13	13	6	$13 - 6 = 7$

For example, **sodium-23** has 23 particles in its nucleus altogether. Eleven are protons, so the remaining **12** are neutrons.

(iv) Solution, suspension or colloid

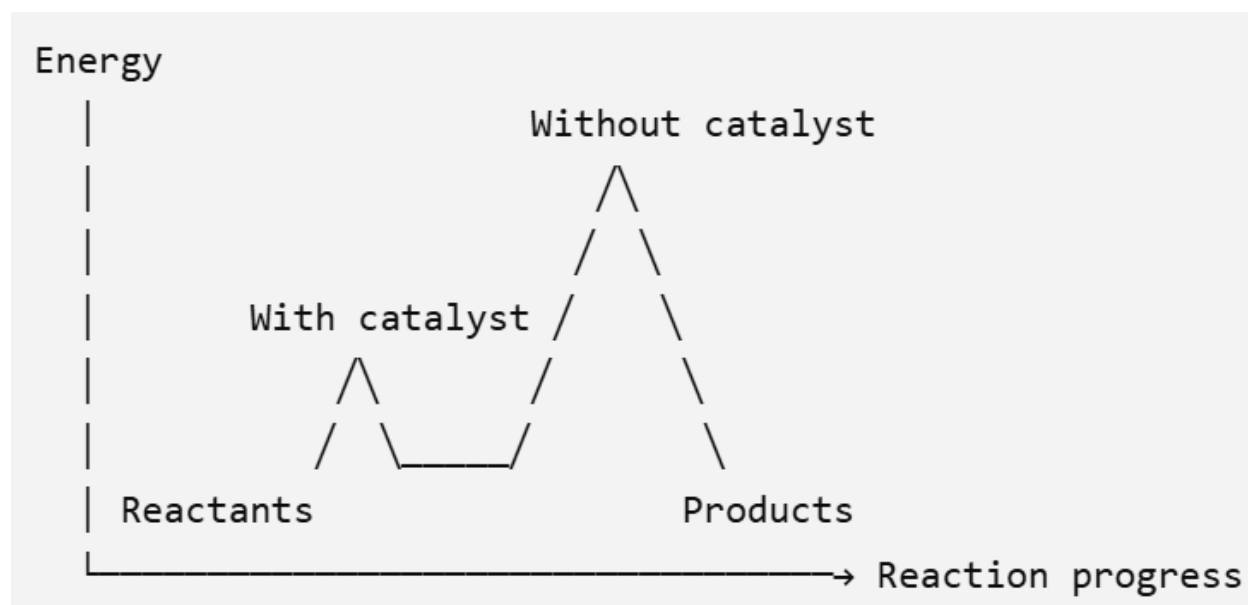
Substance Classification Reason

Air	Solution	A uniform mixture of gases, mainly nitrogen and oxygen; it is a <i>gaseous solution</i> .
Sand in water	Suspension	The relatively large sand particles remain dispersed for a time but eventually settle.
Jelly	Colloid	Liquid is dispersed within a gel structure; its dispersed particles do not settle like sand.

OR: Effect of a catalyst

a. Effect on reaction rate: A catalyst **increases the rate of a chemical reaction** without being consumed in the reaction. It speeds up the formation of products by allowing more reactant molecules to successfully collide and react per unit of time.

b. Effect on activation energy: A catalyst **lowers the activation energy (E_a)** of the reaction. It achieves this by providing an alternative reaction pathway. That alternative route has a **lower activation energy**—a lower energy barrier. At the same temperature, more reacting particles can overcome the barrier, so more successful reactions occur per second.



The sketch illustrates the intended comparison: the **catalyzed route has a lower peak**. A catalyst does not change the identities or overall energy difference of the reactants and products.

(v) Element versus mixture

An **element** is a pure substance made of atoms that all have the **same number of protons**. It cannot be broken down into simpler substances by ordinary chemical means. Examples include iron (Fe) and oxygen (O_2).

One distinguishing feature: An element has **one type of atom**, whereas a mixture contains **two or more substances physically combined** in proportions that can vary. For example, oxygen is an element; air is a mixture.

OR: Formation of an ionic bond

An ionic bond is formed by the **complete transfer of one or more valence electrons** from an electropositive atom (usually a metal) to an electronegative atom (usually a non-metal).

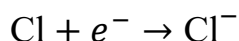
Example: Sodium chloride (NaCl)

Example: Sodium Chloride (NaCl)

- **Sodium (Na):** Electronic configuration is 2,8,1. It loses 1 electron to achieve a stable octet:



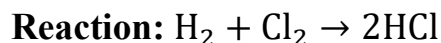
- **Chlorine (Cl):** Electronic configuration is 2,8,7. It gains that 1 electron to achieve a stable octet:



- **Bond Formation:** The electrostatic attraction between Na^+ and Cl^- forms the ionic compound **Sodium Chloride (NaCl)**:



(vi) Enthalpy change from bond energies



Breaking bonds require energy; forming bonds releases energy.

$$\Delta H = \sum (\text{Bond energies of reactants broken}) - \sum (\text{Bond energies of products formed})$$

Bonds broken	Energy
One H–H bond	436 kJ mol ⁻¹
One Cl–Cl bond	242 kJ mol ⁻¹
Total	678 kJ mol⁻¹

Two H–Cl bonds form, releasing $2(431) = 862 \text{ kJ mol}^{-1}$. Therefore,

$$\Delta H = 678 - 862 = -184 \text{ kJ mol}^{-1}$$

Conclusion: The negative sign means the reaction is **exothermic**: it releases about 184 kJ for the reaction as written.

OR: Fluorine compared with chlorine

a. Fluorine has a smaller shielding effect than Chlorine

- **Reason:** The shielding (or screening) effect depends on the number of inner electron shells lying between the nucleus and the valence electrons.
- **Explanation:**
 - **Fluorine (F, atomic number 9):** Electronic configuration is 2,7 (contains 2 **shells**). It has only 1 inner shell ($1s^2$) shielding the valence electrons.
 - **Chlorine (Cl, atomic number 17):** Electronic configuration is 2,8,7 (contains 3 **shells**). It has 2 inner shells shielding the valence electrons.
- **Conclusion:** Since fluorine has fewer inner electron shells than chlorine, it experiences less screening, resulting in a **smaller shielding effect**.

b. Fluorine has higher ionization energy than Chlorine

- **Reason:** Ionization energy depends on atomic radius and effective nuclear charge.
- **Explanation:**
 - Fluorine has a **smaller atomic size** than chlorine because it has fewer electron shells (2 shells vs. 3 shells).
 - Due to the smaller distance between the nucleus and the valence shell, as well as the smaller shielding effect in fluorine, the outer electrons are **more strongly attracted** by the positive nucleus.
- **Conclusion:** More energy is required to remove an electron from a fluorine atom than from a chlorine atom, giving fluorine a **higher ionization energy**.

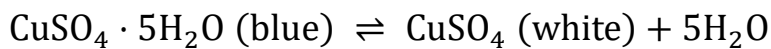
(vii) Effect of temperature on solubility

a. KCl in water: The question states that dissolution is **endothermic**, meaning it absorbs heat. Increasing temperature supplies heat and generally **increases KCl's solubility**.

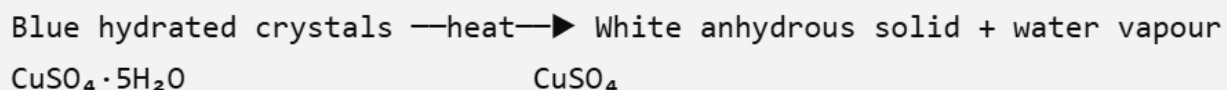
b. Ca(OH)₂ in water: The question states that dissolution is **exothermic**, meaning it releases heat. Increasing temperature generally **decreases its solubility**.

A useful way to remember this is to treat heat as part of the equilibrium: heating favours the direction that **uses up heat**.

OR: Heating hydrated copper(II) sulfate



Heating favours the **forward, dehydration direction**. Water leaves the blue crystals, and **white anhydrous copper(II) sulfate** is formed. If the released water escapes as vapour, its removal further favours the right-hand side.



Adding water to the white solid can hydrate it again, restoring the **blue colour**.

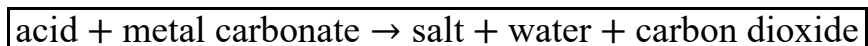
(viii) Metallic character down a group

Trend: Metallic character generally **increases down a group**.

Reason: Each step down adds an electron shell. The outer electrons become farther from the nucleus and experience more shielding by inner electrons. The atom can therefore lose an outer electron more easily. Since the tendency to **lose electrons and form positive ions** is a key metallic property, metallic character increases.

OR: Acid reacting with a carbonate

General reaction:



Example:



Visible change: Bubbles or **effervescence** appear as carbon dioxide gas escapes. If the carbonate is a solid, it may gradually get smaller as it reacts.

(ix) Density and reactivity down Group I

Density: It **generally increases** down Group I as the atoms become heavier. There is an important exception: **potassium is less dense than sodium**, so the trend is not perfectly regular.

Reactivity: It **increases** down Group I. The single outer electron gets farther from the nucleus and is more shielded. It is therefore easier to remove, allowing the metal to form a +1 ion more readily. For example, potassium reacts more vigorously with water than sodium.

OR: Three main types of chemical fertilizers

The three main types supply the nutrients **N, P, and K**:

1. **Nitrogenous fertilizers** — supply nitrogen, such as urea.
2. **Phosphatic fertilizers** — supply phosphorus, such as superphosphate.
3. **Potassic fertilizers** — supply potassium, such as potassium chloride.

(x) Benefits of dissolved oxygen and essential minerals

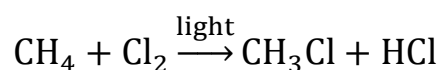
Dissolved oxygen allows fish and many other aquatic organisms to **respire** and release energy from food. If its concentration becomes too low, these organisms can suffocate.

Essential dissolved minerals provide nutrients needed for the growth and normal functioning of aquatic plants and other organisms. For example, plants use suitable amounts of dissolved nitrate and phosphate nutrients to grow; aquatic animals also need minerals for bodily functions. **Excess** nutrients, however, can cause algal blooms that reduce dissolved oxygen.

OR: Substitution reactions

A **substitution reaction** is one in which an atom or group in a molecule is **replaced by another atom or group**.

Alkanes commonly undergo substitution reactions. For example, in the presence of light, a chlorine atom replaces one hydrogen atom in methane:



Here, methane becomes chloromethane because **one H is replaced by Cl**.

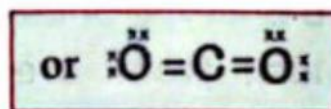
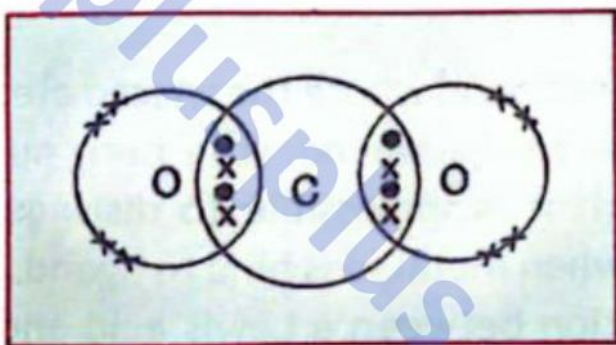
Sec C Long Questions

Q3. Show covalent bond formation using dot-and-cross structures

Key: $\cdot$ and $\times$ are electrons supplied by different atoms. A $\cdot\times$ pair between atoms is **one shared pair**, or one covalent bond. For CO_2 , carbon supplies the crosses; for the other two molecules, the central atom supplies the dots.

(a) Carbon dioxide, CO_2

Carbon has **four valence electrons**; each oxygen has **six**. Carbon shares **two pairs** with each oxygen, forming two double bonds.

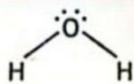
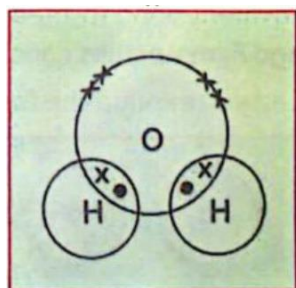


Structural formula: $\boxed{\text{O}=\text{C}=\text{O}}$

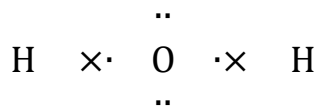
Each oxygen also has **two lone pairs** (the pairs shown above and below it). Carbon and both oxygen atoms complete their octets.

(b) Water, H_2O

Oxygen has **six valence electrons**. It shares one electron with each of two hydrogen atoms, making **two single covalent bonds**. Its remaining four electrons form **two lone pairs**.



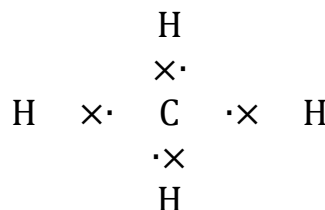
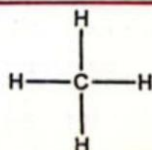
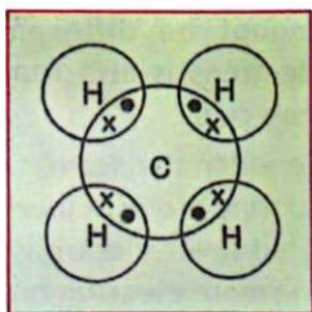
Structural formula: $\boxed{\text{H} - \text{O} - \text{H}}$



Each hydrogen has two electrons in its shared pair; oxygen has eight electrons around it.

(c) Methane, CH_4

Carbon has **four valence electrons** and shares one with each of four hydrogen atoms. This makes **four single covalent bonds**.



Carbon completes its octet, and each hydrogen completes its two-electron outer shell.

Q3 — OR: Molecular and structural formulae

A **molecular formula** gives the **actual number of atoms of each element** in one molecule. For example, the molecular formula $\text{C}_3\text{H}_6\text{O}_2$ tells us that a molecule contains three carbon, six hydrogen and two oxygen atoms. It does **not** show how they are joined.

A **structural formula** shows **how the atoms are connected**. For example, the structural formula of propanoic acid is $\text{CH}_3 - \text{CH}_2 - \text{COOH}$.

The same molecular formula $\text{C}_3\text{H}_6\text{O}_2$ can have different structural formulae:

Compound	Structural formula	Molecular formula
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Propanoic acid	$\text{CH}_3 - \text{CH}_2 - \text{COOH}$	$\text{C}_3\text{H}_6\text{O}_2$
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Methyl ethanoate	$\text{CH}_3 - \text{COO} - \text{CH}_3$	$\text{C}_3\text{H}_6\text{O}_2$
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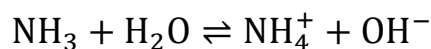
They contain the **same numbers of each type of atom**, but the atoms are **connected differently**. Such compounds are called **structural isomers**.

Q4. Brønsted–Lowry acids and bases

According to the **Brønsted–Lowry theory**:

- An **acid donates a proton**, H^+ .
- A **base accepts a proton**, H^+ .

Consider:



Water **donates** H^+ to ammonia, so H_2O is **the acid** in the forward reaction.

Ammonia **accepts** H^+ , so NH_3 is **the base**.

After donating a proton, water becomes its **conjugate base**, OH^- . After accepting a proton, ammonia becomes its **conjugate acid**, NH_4^+ .

Substance	Role in the forward reaction
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NH_3	Base; accepts H^+
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H_2O	Acid; donates H^+
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NH_4^+	Conjugate acid of NH_3
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OH^-	Conjugate base of H_2O
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Conjugate pairs: $\text{NH}_4^+/\text{NH}_3$ and $\text{H}_2\text{O}/\text{OH}^-$.

Q4 — OR: Three functions each of carbohydrates and proteins

Carbohydrates

Immediate energy: Glucose is broken down in cellular respiration to release energy for body activities.

Energy storage: Excess glucose can be stored as **glycogen** in the liver and muscles for later use.

Digestive support: Dietary fibre, a form of carbohydrate, helps food move through the digestive system and supports regular bowel movements.

Proteins

Growth and repair: Proteins provide amino acids needed to build and repair tissues, including muscles.

Enzymes: Many enzymes are proteins that speed up essential chemical reactions in the body.

Defence: Antibodies are proteins that help the immune system recognise and fight harmful microbes.

Q5. Compare metals and non-metals

Property	Metals	Non-metals
(a) Thermal conductivity	Generally good conductors of heat ; for example, copper transfers heat readily.	Generally poor conductors of heat ; many act as insulators.
(b) Electrical conductivity	Generally good conductors of electricity because mobile electrons can carry charge.	Generally poor conductors of electricity because they usually lack mobile charge carriers. Graphite is an important exception.
(c) Malleability and ductility	Generally malleable (can be hammered into sheets)	Solid non-metals are generally brittle : they break instead of forming sheets or wires.

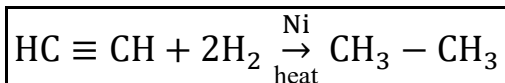
Property	Metals	Non-metals
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and **ductile** (can be drawn into wires).

These are **general trends**, so state the graphite exception when discussing electrical conduction.

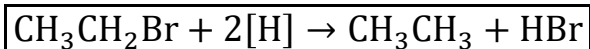
Q5 — OR: Preparation of ethane

(a) By hydrogenation of an alkyne: Ethyne, C_2H_2 , reacts with **two molecules of hydrogen** in the presence of a nickel catalyst to form ethane. Its triple bond is fully hydrogenated to a single bond.



The product $CH_3 - CH_3$ is **ethane**. Adding only one molecule of hydrogen would give **ethene**, so $2H_2$ is needed for complete hydrogenation to ethane.

(b) By reduction of an alkyl halide: An **ethyl halide**, such as bromoethane, is reduced. The halogen atom is replaced by hydrogen:



Here $[H]$ represents hydrogen supplied by a suitable reducing system. The starting halide must have **two carbon atoms** to produce the two-carbon alkane **ethane**.

FBISE Class 9th Chemistry Solution 1st Annual Exam, 2026

Code 12051

Sec A MCQs

1. Which state of matter is air? C — Gas

Air is a **mixture of gases**, mainly nitrogen and oxygen. It spreads out to fill the space available.

- **A, solid:** A solid has a fixed shape; air does not.
- **B, liquid:** A liquid has a fixed volume; air expands to fill its container.
- **D, plasma:** Plasma contains charged particles. Ordinary air is not plasma.

2. Diamond and graphite are forms of which element? B — Carbon

Diamond and graphite are both made entirely of **carbon atoms**. Their atoms are arranged differently, giving them different properties. Different forms of the same element are called *allotropes*.

A (oxygen), C (silicon), and D (sulfur) are different elements; neither diamond nor graphite is made of them.

3. Sodium changes from oxidation state 0 to +1. What process is this? A — Oxidation

In $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$, each sodium atom **loses one electron** to form Na^+ . Its oxidation state rises from 0 to +1. Losing electrons, or an increase in oxidation state, is **oxidation**.

- **B, reduction:** Reduction involves gaining electrons or a *decrease* in oxidation state.
- **C, elimination:** This does not describe sodium's electron loss.
- **D, addition:** This does not describe sodium's change in oxidation state.

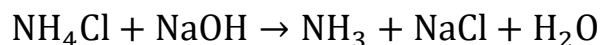
4. What is a reaction that does not go to completion called? C — Reversible

In a **reversible reaction**, products can change back into reactants. The forward and backward reactions can reach equilibrium before all the reactants are used up.

- **A, irreversible:** This generally proceeds in one direction under the stated conditions.
- **B, endothermic:** This means a process absorbs heat; it does not say whether it is reversible.
- **D, exothermic:** This means a process releases heat; it does not say whether it is reversible.

5. What is produced when a base reacts with an ammonium salt? D — Ammonia

A base removes a hydrogen ion from the ammonium ion, NH_4^+ , producing **ammonia**, NH_3 . For example:



A (hydrogen), B (carbon dioxide), and C (oxygen) are not the characteristic gas produced by this reaction.

6. What determines the charge on an ion? C — Valence electrons

Valence electrons are the outermost electrons. An atom becomes an ion when it **loses or gains electrons**: losing one gives a +1 charge; gaining one gives a -1 charge.

- **A, neutrons:** They have no electric charge.
- **B, protons:** Protons are positively charged, but their number stays the same when an atom forms an ordinary ion.
- **D, atomic mass:** Mass does not determine the ion's charge.

More precise rule: An ion's charge equals its **number of protons minus its number of electrons**. Among the given options, **C** is intended because gaining or losing valence electrons changes that difference.

7. Incorrect halogen reactivity relationship — B: $\text{Br}_2 > \text{F}_2$

Fluorine is **more reactive** than bromine, so the correct relationship is $F_2 > Br_2$. This makes **B** the requested *incorrect* statement.

- **A**, $F_2 > Cl_2$: Correct; fluorine is above chlorine in the group.
- **C**, $Br_2 > I_2$: Correct; bromine is above iodine.
- **D**, $Cl_2 > Br_2$: Correct; chlorine is above bromine.

A more reactive halogen can displace a less reactive halogen from its halide solution.

8. Percentage of oxygen in air — A: 21%

About **21%** of clean, dry air is oxygen. **B (50%)** and **D (92%)** are far too high. **C (78%)** is approximately the percentage of **nitrogen**, not oxygen.

9. Factor increasing water scarcity — A: Population growth

As the population grows, more water is needed for homes, food production, and industry. If water supplies do not grow at the same rate, scarcity increases.

- **B, cloud formation:** Clouds are part of the water cycle; their formation alone does not establish increased scarcity.
- **C, wind speed:** Wind can affect evaporation, but it is not the principal factor this question is testing.
- **D, soil erosion:** It can harm water storage and quality, so it may **contribute indirectly** to water problems. Population growth is the clearest intended answer here.

10. General formula of carboxylic acids — B and C both work

This MCQ has **two chemically valid options** for *saturated monocarboxylic acids*:

- **B**, $C_nH_{2n}O_2$: Here n counts **all carbon atoms** in the acid. For ethanoic acid, $n = 2$, giving $C_2H_4O_2$.
- **C**, $C_nH_{2n+1}COOH$: Here n counts the carbon atoms in the **alkyl group before** $COOH$. For ethanoic acid, $n = 1$, giving CH_3COOH —the same $C_2H_4O_2$.

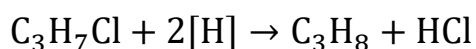
A, $C_nH_{2n+2}COOH$, has too many hydrogens in the alkyl part. **D**, $C_nH_{2n}COOH$, does not give the usual saturated alkyl-group formula. If a marking scheme requires one choice, check which form the textbook specifies; **the question as printed does not have a unique correct answer.**

11. Fish is a good source of — C: Protein

Fish provides **protein**, which the body uses for growth and repair. **A (carbohydrate)** is a different nutrient category; **B (sugar)** and **D (glucose)** are types of carbohydrate, not the main nutrient intended here.

12. Reduction of propyl chloride — B: Propane

Propyl chloride has three carbon atoms. In this reduction, its chlorine is replaced by hydrogen, producing the three-carbon alkane **propane**:



- **A, propene:** Has a carbon–carbon double bond; it can form through *elimination*, rather than the stated reduction.
- **C, methane:** Has only one carbon atom; reduction does not remove two carbons here.
- **D, propyne:** Has a triple bond; that is not produced by this reduction.

Sec B Short Questions

(i) Identify the branch of chemistry

a. Boyle's law: pressure and volume of a gas — Physical chemistry. Physical chemistry studies the physical behaviour of substances and the laws governing it. Boyle's law describes how a gas's volume decreases when its pressure increases at constant temperature.

b. Industrial SO_2 causing acid rain — Environmental chemistry. This branch studies chemicals released into the environment and their effects on air, water and living things. Sulfur dioxide can contribute to acid rain.

c. Keratin protein in hair — Biochemistry. Biochemistry studies the substances and chemical processes of living organisms. Keratin is a protein found in hair.

OR: Suspension versus colloid

Property	Suspension	Colloid
Dispersed particles	Relatively large	Smaller than suspension particles
On standing	Particles usually settle	Particles do not normally settle
Separation	Often separable by ordinary filtration	Not normally separable by ordinary filtration
Example	Sand in water	Milk

Exam answer: Sand in water is a suspension because the sand eventually settles. Milk is a colloid because tiny droplets of fat remain dispersed in water without quickly settling.

(ii) Energy and distance as shell number increases

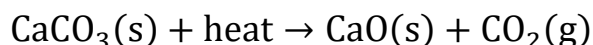
As the **shell number increases**, the electron shell lies **farther from the nucleus**. For example, the third shell is farther out than the second.

The **energy of electrons in the higher shell also increases**. Such electrons are held less tightly by the nucleus because they are farther away and inner electrons shield them. “Higher energy” here means the electron is **less strongly bound**; it does not mean more energy is needed to remove it.

OR: Reaction with a positive enthalpy change

An **endothermic reaction** has a **positive enthalpy change**, $\Delta H > 0$. It **absorbs heat from its surroundings**, so the products have higher enthalpy than the reactants.

Example: Thermal decomposition of calcium carbonate



Heat must be supplied for this reaction; therefore its ΔH is positive.

(iii) Particle determining atomic number

The **number of protons** in an atom’s nucleus determines its **atomic number**, Z .

$Z = \text{number of protons}$

Significance: The atomic number uniquely **identifies an element**. Every carbon atom, for example, has six protons, so carbon’s atomic number is 6. Changing the number of protons changes the element. For a **neutral atom**, atomic number also equals the number of electrons; this helps determine its electron arrangement and position in the periodic table.

OR: Melting point and density down Group I-A

Melting point generally decreases down Group I-A. The metal atoms become larger, and the attraction holding them together in the metallic structure becomes weaker; less heat is needed to melt the metal.

Density generally increases down the group as the atoms become heavier. This trend has an **exception**: potassium is less dense than sodium. Thus, write “**generally increases**,” not “always increases.”

(iv) Justify the statements

a. An atom is electrically neutral. An atom contains **positively charged protons** and **negatively charged electrons**. In a neutral atom, their numbers are equal, so the positive and negative charges cancel. Neutrons have no charge.

b. An atom becomes a positive ion when it loses an electron. Losing an electron leaves **more protons than electrons**. Since protons carry positive charge, the particle has an overall positive charge. For example:



A sodium atom that loses one electron becomes a +1sodium ion.

OR: Activation energy and the effect of a catalyst

Activation energy is the **minimum energy reacting particles must have for a successful reaction**. It is the energy barrier between reactants and products.

A catalyst provides an alternative reaction pathway with lower activation energy. At the same temperature, more particles can overcome this lower barrier, so the reaction proceeds faster. The catalyst does not change the overall enthalpy change of the reaction.

(v) Find the mass of 0.5 mole of NaCl

The molar mass of sodium chloride is:

$$M(\text{NaCl}) = 23 + 35.5 = 58.5 \text{ g mol}^{-1}$$

Using $m = nM$:

$$m = 0.5 \text{ mol} \times 58.5 \text{ g mol}^{-1} = \boxed{29.25 \text{ g}}$$

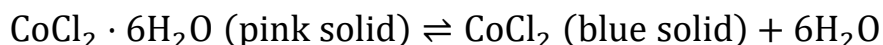
OR: Derive the ionic formulae

An ionic compound has **zero overall charge**. Choose enough of each ion to balance the positive and negative charges.

Ions	Charge balance	Formula
a. Mg^{2+} , Cl^-	One $+2$ ion needs two -1 ions	MgCl_2
b. Al^{3+} , SO_4^{2-}	Two Al ions give $+6$; three sulfate ions give -6	$\text{Al}_2(\text{SO}_4)_3$
c. Ca^{2+} , NO_3^-	One $+2$ ion needs two -1 ions	$\text{Ca}(\text{NO}_3)_2$

Why the brackets? SO_4^{2-} and NO_3^- are *whole polyatomic ions*. Brackets show that the subscript applies to the entire ion.

(vi) Add water to the cobalt chloride equilibrium

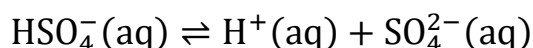
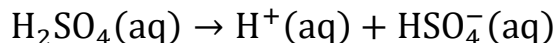


Adding water increases the amount of a substance on the **right-hand side**. By Le Châtelier's principle, the equilibrium shifts **to the left** to use some of the added water. Blue anhydrous cobalt chloride becomes **pink hydrated cobalt chloride**.

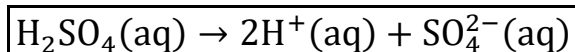
Addition of water: equilibrium shifts left; blue solid turns pink.

OR: Dissociation equations in water

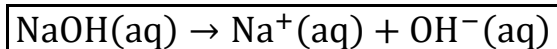
Sulfuric acid releases hydrogen ions in two stages:



For the school-level **overall** equation:



Sodium hydroxide dissociates into sodium and hydroxide ions:



The second ionization of sulfuric acid is shown separately as reversible because it is less complete than the first.

(vii) Electronic configuration 2, 8, 7

The atom has electrons in **three occupied shells**, so it is in **Period 3**. It has **seven valence electrons**, so it belongs to **Group 17 (Group VII-A)**. The element with atomic number 17 is **chlorine**.

Requested property Answer

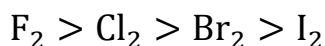
Period **3**

Group **17 / VII-A**

Valence electrons **7**

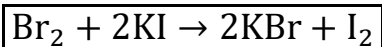
OR: Complete the halogen displacement reactions

Halogen reactivity decreases down Group 17:

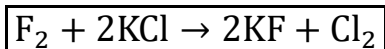


A more reactive halogen displaces a less reactive one from its halide.

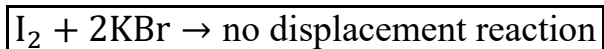
a. Bromine and potassium iodide: Bromine is more reactive than iodine, so displacement occurs.



b. Fluorine and potassium chloride: Fluorine is more reactive than chlorine, so the expected displacement equation is:



c. Iodine and potassium bromide: Iodine is less reactive than bromine, so it cannot displace bromine.



Exam note: The equation in **b** expresses the halogen reactivity rule. In actual aqueous solution, fluorine can also react with water, so real conditions are more complicated.

(viii) Benefits of dissolved oxygen and minerals in water

Dissolved oxygen is used by fish and many other aquatic animals for **respiration**. It helps them release energy from food and survive.

Essential dissolved minerals provide nutrients needed by aquatic plants and other organisms for **growth and normal body functions**. For example, suitable amounts of mineral nutrients support plant growth, which helps sustain aquatic food chains.

OR: Classify the compounds

Formula	Name	Classification	Identifying feature
a. $\text{CH}_3\text{OCH}_2\text{CH}_3$	Methoxyethane	Ether	An oxygen atom joins two carbon groups: $\text{R} - \text{O} - \text{R}'$.
b. CH_3OH	Methanol	Alcohol	An $-\text{OH}$ group is attached to a non-aromatic carbon.
c. $\text{C}_6\text{H}_5\text{OH}$	Phenol	Phenol	An $-\text{OH}$ group is attached directly to a benzene ring.

(ix) Sedimentation and chlorination in water treatment

Sedimentation: Water is left undisturbed so heavier suspended particles, such as sand and mud, **settle to the bottom**. The clearer water can then be separated from the sediment.

Chlorination: A controlled amount of chlorine is added to **kill or inactivate disease-causing microorganisms**, making the treated water safer to drink. Settling particles and disinfecting water serve different purposes, so both steps are useful.

OR: Homologous series

A **homologous series** is a family of organic compounds that have the **same functional group and general formula**, with successive members differing by a

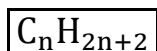
—CH₂—unit. For example, methane (CH₄) and ethane (C₂H₆) are successive members of the alkane series.

Members have **similar chemical properties** because they contain the **same functional group or characteristic bonding pattern** that takes part in their reactions. Their physical properties, such as boiling point, may change gradually as molecular size increases.

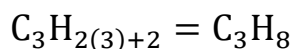
(x) Why are alkanes saturated?

Alkanes contain **only single carbon–carbon bonds**. Each carbon therefore has the maximum number of hydrogen atoms possible for that carbon arrangement. There is no carbon–carbon double or triple bond, so alkanes are called **saturated hydrocarbons**.

For a non-cyclic alkane:



Here n is the number of carbon atoms. For example, if $n = 3$:



Thus, the molecular formula of propane is C₃H₈.

OR: Nutrition and healthy eating

Nutrition is the process by which the body obtains food, digests and absorbs nutrients, and uses them for energy, growth, repair and health.

Healthy eating provides an appropriate variety and amount of nutrients. It is important because it:

1. Supplies **energy** for daily activities.
2. Supports **growth and repair** of body tissues.
3. Helps maintain normal body functions and prevents **nutrient deficiencies**.

Sec C Long Questions

Q3. Differentiate between compounds and mixtures

Point	Compound	Mixture
a. Composition	Elements combine in a fixed ratio . Water is always H ₂ O: two hydrogen atoms for each oxygen atom.	Substances can be present in varying proportions . The amounts of salt and water in salt water can change.
b. Chemical bonding	The constituent atoms are chemically bonded , forming a new substance with its own properties.	Components are physically mixed ; they do not have to form new chemical bonds and generally retain their own properties.
c. Separation	Components require a chemical change to separate them into simpler substances.	Components can generally be separated by physical methods , such as filtration, distillation or evaporation.

Example: Water is a compound; salt water is a mixture.

Q3 — OR: Importance of lipids

Lipids include fats and oils. A complete answer gives **three specific sources and three functions**.

Sources of dietary lipids	Functions in the body
Vegetable oils , such as olive or sunflower oil	Long-term energy storage: Fat stores energy that the body can use later.
Nuts and seeds , such as walnuts and sunflower seeds	Cell membranes: Lipids form a major part of the membranes surrounding cells.

Sources of dietary lipids Functions in the body

Fatty fish , such as salmon or sardines	Insulation and protection: Fat beneath the skin reduces heat loss, while fat around some organs cushions them.
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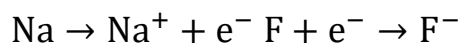
Lipids also help the body absorb the fat-soluble vitamins **A, D, E and K**.

Q4. Ionic bond formation using dot-and-cross structures

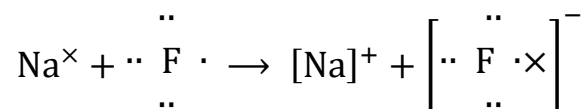
Key: In the diagrams, $\cdot$ represents an electron originally belonging to the non-metal, while $\times$ represents an electron transferred from the metal. The completed outer shell of each negative ion has **eight electrons**.

(a) Sodium fluoride, NaF

Sodium has **one valence electron** (2, 8, 1); fluorine has **seven** (2, 7). Sodium transfers its electron to fluorine:



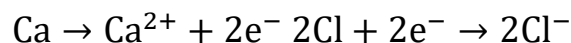
Electron-dot-and-cross representation of the resulting ions:



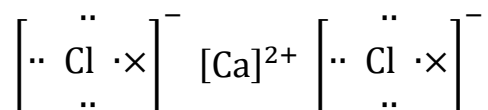
Fluorine's seven original dots plus sodium's one cross make **eight outer electrons**. The electrostatic attraction between Na^+ and F^- forms NaF.

(b) Calcium chloride, CaCl_2

Calcium has **two valence electrons** (2, 8, 8, 2). Each chlorine atom has **seven** (2, 8, 7). Calcium gives **one electron to each of two chlorine atoms**:



A compact dot-and-cross picture of the ions is:



Charge check: $+2 + (-1) + (-1) = 0$. Thus one calcium ion and two chloride ions form CaCl_2 . The bond is the **electrostatic attraction between the oppositely charged ions**, not the electron transfer itself.

Q4 — OR: Element Z below magnesium in Group II-A

Going **down Group II-A**, atoms gain an electron shell. Their two outer electrons are farther from the nucleus and experience greater shielding.

a. Reactivity: Z is **more reactive** than magnesium because it can lose its two outer electrons more easily to form Z^{2+} .

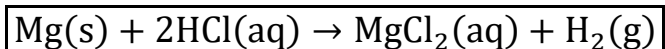
b. Metallic character: Z has **greater metallic character** than magnesium. Metallic character reflects how readily an atom loses electrons.

c. Ionization energy: Z has a **lower first ionization energy** than magnesium because less energy is required to remove an outer electron that is farther from, and more shielded from, the nucleus.

The element immediately below magnesium is **calcium**; the same trend applies to elements farther down the group.

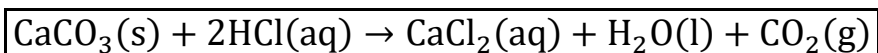
Q5. Reactions of hydrochloric acid

(a) With magnesium



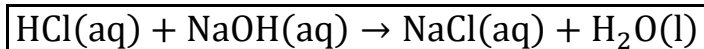
Reaction type: Acid + metal $\rightarrow$ salt + hydrogen. **Observation:** Bubbles of hydrogen gas appear as magnesium dissolves.

(b) With calcium carbonate



Reaction type: Acid + carbonate $\rightarrow$ salt + water + carbon dioxide. **Observation:** Effervescence occurs as CO_2 escapes. The gas turns limewater milky.

(c) With sodium hydroxide



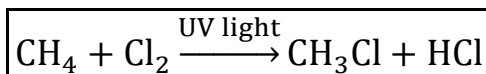
Reaction type: Neutralization. The acid and base form **salt and water**; the reaction usually releases heat.

Q5 — OR: Substitution in alkanes

A **substitution reaction** replaces one atom or group in a molecule with another atom or group.

Alkanes are **saturated**: they have only single carbon–carbon bonds, so they do not normally undergo the addition reactions characteristic of alkenes. Under suitable conditions, a halogen can instead **replace a hydrogen atom** of an alkane.

In **mono substitution** of methane with chlorine, **one** of methane's four hydrogens is replaced by chlorine in the presence of sunlight or ultraviolet light:



The monosubstituted organic product is **chloromethane**, CH_3Cl . If chlorination continues, more hydrogens can be replaced; the equation above shows the requested **single substitution**.

FBISE Class 9th Chemistry Solution 1st Annual Exam, 2026

Code 19051

Sec A MCQs

1. Where are protons and neutrons? C — Nucleus

The **nucleus** is the small, central part of an atom. It contains protons and neutrons.

- **A, electron cloud:** Describes the region where electrons are found.
- **B, shells:** Describe electron energy levels, not where protons and neutrons are found.
- **D, orbits:** In a simplified atomic model, these describe electrons moving around the nucleus.

2. Which compounds generally have low boiling points? D — Covalent compounds

Many **simple molecular covalent compounds** have weak attractions *between their molecules*. Relatively little heat is needed to separate those molecules, so their boiling points are often low.

- **A, ionic compounds:** Strong attraction between positive and negative ions usually gives high boiling points.
- **B, network solids:** Strong covalent bonds extend throughout the structure, giving very high boiling points.
- **C, metallic compounds:** Metallic bonding generally gives comparatively high boiling points.

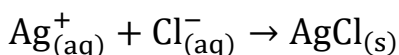
Important distinction: “Covalent” does not *always* mean low boiling point. Network covalent substances are an exception. **D** refers to the usual *simple molecular* covalent compounds.

3. Formula formed by Al^{3+} and O^{2-} — C: Al_2O_3

A compound must have **zero overall charge**. Two aluminum ions give $2(+3) = +6$, while three oxide ions give $3(-2) = -6$. The charges cancel, giving Al_2O_3 .

- **A, AlO :** $+3 - 2 = +1$; charges do not cancel.
- **B, Al_2O :** $+6 - 2 = +4$; charges do not cancel.
- **D, Al_3O_2 :** $+9 - 4 = +5$; charges do not cancel.

4. Net ionic equation for silver chloride — C



Dissolved silver and chloride ions combine to form **solid silver chloride**, a precipitate. A *net ionic equation* shows only the particles that take part in that change.

- **A:** Shows sodium chloride, not silver chloride.
- **B:** Is a full chemical equation; it also includes Na^+ and NO_3^- , which are spectator ions.
- **D:** Starts with solid silver and chlorine gas rather than the dissolved ions involved in precipitation.

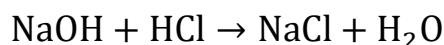
5. Symbol for a reversible reaction — B: $\rightleftharpoons$

The two opposing arrows show that the reaction can proceed **forward** (reactants to products) and **backward** (products to reactants).

- **A, $\rightarrow$:** Shows a reaction written in one direction.
- **C, $=$:** Is an equals sign, not a reaction arrow.
- **D, $\leftrightarrow$:** Can suggest movement both ways in general diagrams, but the **standard chemical reaction/equilibrium notation intended here is B**. In chemistry, $\leftrightarrow$ is also used for resonance structures, which are not two substances reacting back and forth.

6. Which reaction shows a property of a base? D — **Base + acid $\rightarrow$ salt + water**

A base **neutralizes an acid**. The usual products are a salt and water. For example:



- **A, base + metal → salt:** This is not the general neutralization reaction.
- **B, base + water → acid:** Adding water to a base does not turn it into an acid.
- **C, base → ions only:** Some bases form ions in water, but “ions only” does not describe the products of the acid–base reaction asked about here.

7. What determines an element's period number? A — Occupied shells

The **period number** tells us how many electron shells contain electrons. For example, sodium has electrons in three shells, so it is in Period 3.

- **B, electrons:** The *total* number of electrons does not directly give the period; their arrangement into shells does.
- **C, protons:** Their number identifies the element (its atomic number), not directly its period.
- **D, neutrons:** These do not determine the period.

8. What mainly produces carbon monoxide and particulates? B — Incomplete combustion

When fuel burns with **insufficient oxygen**, some carbon forms poisonous carbon monoxide (CO) and fine particles such as soot instead of being fully converted to carbon dioxide.

- **A, complete combustion:** With enough oxygen, carbon mainly forms CO₂.
- **C, plant respiration:** Does not produce soot or carbon monoxide as described.
- **D, evaporation:** A physical change, not burning.

9. Which acid links sulfur to acid rain? D — H₂SO₄

Burning sulfur-containing fuels releases sulfur dioxide. It can be further oxidized and react with water in the atmosphere to form **sulfuric acid**, H₂SO₄, a contributor to acid rain.

- **A, HNO₃:** Nitric acid is linked mainly to nitrogen oxides.
- **B, H₂CO₃:** Carbonic acid forms when carbon dioxide dissolves in water.
- **C, HCl:** Hydrochloric acid is not the acid formed from sulfur oxides.

10. What in natural water is useful for aquatic life? **A — Dissolved oxygen**

Fish and many other aquatic animals need **oxygen dissolved in water** for respiration.

- **B, sewage:** Pollutes water and can reduce its dissolved oxygen.
- **C, dissolved nitrates:** Small amounts are nutrients for plants, but excess nitrates can cause algal growth that ultimately harms aquatic life. **Oxygen is the clearest answer to this question.**
- **D, dissolved plastics:** Plastics are pollutants, not a useful dissolved gas or nutrient.

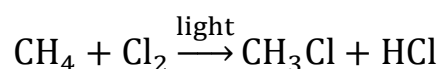
11. Which compound is an alkane? **B — C₄H₁₀**

Alkanes have only single carbon–carbon bonds and follow the formula C_nH_{2n+2}. With $n = 4$, this gives C₄H₁₀, called **butane**.

- **A, C₂H₄:** Ethene, an alkene with a double bond.
- **C, C₂H₂:** Ethyne, an alkyne with a triple bond.
- **D, C₃H₆:** Does not match the alkane formula; propane would be C₃H₈.

12. Where do substitution reactions commonly occur? **A — Alkanes**

In a **substitution reaction**, one atom is replaced by another. For example, light can help chlorine replace a hydrogen atom in methane:



- **B, alkynes** and **C, alkenes:** Their multiple bonds commonly undergo *addition* reactions.
- **D, aldehydes:** These contain a –CHO group and are not the class associated with the common alkane substitution reaction tested here.

Sec B Short Questions

(i) Identify the branch of chemistry

a. Rising Earth temperature associated with increasing CO₂— Environmental chemistry. This branch studies chemicals in the environment and their effects. Carbon dioxide is a greenhouse gas; increasing its atmospheric concentration contributes to warming.

b. Large-scale manufacture of sulfuric acid by the Contact process — Industrial chemistry. Industrial chemistry concerns the manufacture of chemicals on a large scale.

c. Cooling a gas and its change of state — Physical chemistry. Physical chemistry studies physical properties and changes of substances, including the behaviour of gases and changes of state.

Correction to statement (c): The printed claim that *a gas liquefies when cooled to –273°C* is scientifically inaccurate. Approximately –273.15°C is **absolute zero**; a gas's liquefaction temperature depends on the **substance and pressure**, and it may liquefy well before that temperature. The intended **branch** is still physical chemistry.

OR: Effect of increasing temperature on solubility

Substance	Dissolving process given	Effect of increasing temperature
KCl in water	Endothermic: absorbs heat	Solubility increases . Added heat favours dissolving.
Ca(OH)₂ in water	Exothermic: releases heat	Solubility decreases . Heating favours the reverse direction.

This follows Le Châtelier's principle: adding heat favours whichever direction **uses up heat**.

(ii) Paths of positive and negative charges in a uniform electric field

The electric field points **from the positive plate to the negative plate**. The force on a charge is $F = qE$:

- A **positive charge** feels a force **in the direction of the field**, towards the **negative plate**.
- A **negative charge** feels a force **opposite to the field**, towards the **positive plate**.

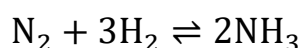
Here is a sketch for particles entering **from the left with horizontal velocity**:

Positive plate (+)
negative particle: $\rightarrow \nearrow$ force upward
electric field: $\downarrow \downarrow \downarrow$
positive particle: $\rightarrow \searrow$ force downward
Negative plate (-)

With horizontal initial motion, their paths **curve in opposite directions**. If released **from rest**, they move along the field line in opposite directions instead.

OR: Reversible reaction and completion

A **reversible reaction** can proceed in **both directions**: reactants form products, and products can react to form reactants again. It is represented by opposing reaction arrows, for example:



In a **closed system under conditions where both directions occur**, it generally **does not go to completion**. As products build up, the backward reaction increases. Eventually, **dynamic equilibrium** may be reached: the forward and backward reaction rates are equal, so the amounts remain constant even though both reactions continue.

(iii) Using carbon-14 to determine age

Carbon-14 dating estimates the age of material that was once part of a living organism, such as wood or bone.

1. While alive, the organism continually takes in carbon, including a small amount of radioactive **carbon-14**.
2. After it dies, carbon intake stops. The carbon-14 already present **gradually decays**.
3. Scientists measure how much carbon-14 remains and compare it with the expected amount in living material. **Less remaining carbon-14 indicates more time since death.**

Carbon-14 has a **half-life of about 5,730 years**: after that time, roughly half the original amount remains. This method dates **once-living material**, rather than the age of any ancient object regardless of what it is made from.

OR: Molecules in 2 moles of CO₂

One mole contains Avogadro's number, 6.022×10^{23} , of molecules.

$$N = nN_A = 2(6.022 \times 10^{23}) = \boxed{1.2044 \times 10^{24} \text{ molecules of CO}_2}$$

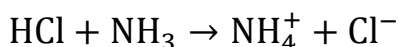
(iv) Activation energy and successful collision

Activation energy is the **minimum energy barrier** that reacting particles must overcome for a reaction to occur.

A **successful collision** is a collision that **produces a chemical reaction**. For a collision to be successful, particles must collide with **enough energy to overcome the activation-energy barrier** and with a **suitable orientation** so the required bonds can break and form. Not every collision meets these conditions.

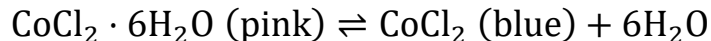
OR: Brønsted–Lowry concept with HCl and NH₃

A **Brønsted–Lowry acid** donates a proton, H⁺. A **Brønsted–Lowry base** accepts a proton.



Here **HCl is the acid** because it donates H⁺. **NH₃ is the base** because it accepts H⁺, becoming NH₄⁺. The ions may combine to give ammonium chloride, NH₄Cl.

(v) Effect of adding water to cobalt chloride

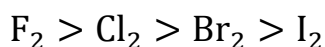


Water is on the **right-hand side**. Adding water shifts the equilibrium **to the left** to use some of it, according to Le Châtelier's principle. Blue anhydrous cobalt chloride takes up water and forms **pink hydrated cobalt chloride**.

Answer: The blue solid turns pink as more $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ forms.

OR: Halogen displacement

Halogen reactivity decreases down the group:



A more reactive halogen can displace a less reactive halogen from its halide.

Part	Reaction	Result
a	$\text{Cl}_2 + 2\text{KF}$	No displacement: chlorine cannot displace the more reactive fluorine.
b	$\text{F}_2 + 2\text{KI} \rightarrow 2\text{KF} + \text{I}_2$	Occurs: fluorine displaces iodine.
c	$\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$	Occurs: chlorine displaces bromine.

These are the expected equations for applying the reactivity series. In aqueous solution, fluorine can also react with water, so part **b** is an idealized displacement equation.

(vi) Modern periodic table

The **periodic table** is a systematic arrangement of chemical elements in rows called **periods** and columns called **groups**. Elements with similar chemical properties generally appear in the same group.

In the **modern periodic table**, elements are arranged in order of **increasing atomic number**—that is, increasing **number of protons**. For example, magnesium

has atomic number 12 and aluminium has atomic number 13, so magnesium comes first.

OR: Three nitrogen-containing fertilizers

Any three suitable examples are:

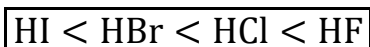
1. **Urea:** $\text{CO}(\text{NH}_2)_2$
2. **Ammonium nitrate:** NH_4NO_3
3. **Ammonium sulfate:** $(\text{NH}_4)_2\text{SO}_4$

Each supplies **nitrogen**, a nutrient plants need for growth.

(vii) Thermal stability of hydrogen halides

Thermal stability means how well a compound **resists decomposition when heated**. A more thermally stable hydrogen halide is harder to break down by heat.

Down the halogen group, the hydrogen–halogen bond generally becomes **longer and weaker**. The H–F bond is strongest and the H–I bond is weakest. Therefore, the order of **increasing thermal stability** is:



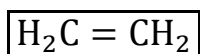
OR: Healthy eating

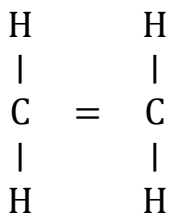
Healthy eating means consuming a **balanced variety of foods in suitable amounts** so the body receives enough nutrients and energy without regularly consuming excessive amounts.

It supports body functions by providing **energy** for activity, **proteins and minerals** for growth and tissue repair, and **vitamins and other nutrients** needed for normal processes and good health.

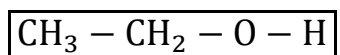
(viii) Structural formulae of ethene and ethanol

Ethene, C_2H_4 : The two carbon atoms have a **double bond**. Each carbon is also bonded to two hydrogen atoms.





Ethanol, C₂H₅OH: The carbon atoms have a **single bond**, and an –OH group is attached to the second carbon.



In ethanol, the O – H bond is essential: the oxygen is bonded to both carbon and hydrogen.

OR: Three main biomolecules and one food source each

Biomolecule	One food source
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Carbohydrates	Rice
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Proteins	Eggs
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Lipids (fats and oils)	Olive oil
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(ix) Three characteristics of a homologous series

1. Its members have the **same functional group** and follow the **same general formula**.
2. Each successive member differs from the previous one by a **CH₂ unit**. For example, ethane C₂H₆ follows methane CH₄.
3. Members have **similar chemical properties**, while physical properties such as boiling point change **gradually** with increasing molecular size.

OR: Carbohydrate-rich foods and their role

Examples: Rice and bread are carbohydrate-rich foods.

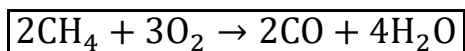
Carbohydrates are a major **source of energy**. During digestion, many carbohydrates are broken down into glucose, which cells use in respiration to

release energy. Excess glucose can be stored as **glycogen** in the liver and muscles for later use.

(x) Combustion of methane

(a) Limited oxygen supply

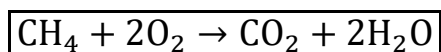
With insufficient oxygen, methane undergoes **incomplete combustion**. One possible product is poisonous **carbon monoxide**:



With even less oxygen, **soot (carbon)** may also form. Thus, limited oxygen does not guarantee only one set of products.

(b) Excess oxygen supply

With sufficient oxygen, methane undergoes **complete combustion** to form carbon dioxide and water:



Both equations are balanced for carbon, hydrogen and oxygen atoms.

OR: Compressibility of solids, liquids and gases

State	Compressibility	Reason
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Solid	Very low	Particles are closely packed, with almost no empty space to remove.
Liquid	Low	Particles are close together, although they can move past one another.
Gas	High	Particles are far apart; pressure can push them much closer together.

Sec C Long Questions

Q3. Compare diamond and graphite

Diamond and graphite are **allotropes of carbon**: both consist only of carbon atoms, but their atoms are bonded in different arrangements.

Feature	Diamond	Graphite
Structure	Each carbon atom forms four covalent bonds in a rigid, three-dimensional network.	Each carbon atom forms three covalent bonds in a flat, hexagonal layer. The layers are held together by weak attractions.
Electrons and conductivity	All four outer electrons take part in bonds, so there are no mobile electrons; diamond does not conduct electricity .	The fourth outer electron is delocalized within a layer; graphite conducts electricity .
Hardness	Very hard because strong bonds extend throughout the structure.	Soft and slippery because its layers can slide over one another.
Melting behaviour	Has a very high melting/sublimation temperature due to strong covalent bonding.	Also requires very high temperature to break its strong bonds within layers.

Key structural picture: Diamond is a **three-dimensional bonded network**. Graphite consists of **stacked bonded sheets** that can slide.

Q3 — OR: Metallic bonding and properties

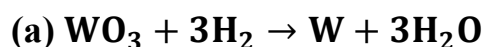
In a metal, outer electrons become **delocalized**: they can move throughout the structure. This leaves a regular arrangement of **positive metal ions**. A **metallic bond** is the **electrostatic attraction between these positive ions and the surrounding delocalized electrons**. The attraction holds the metal together.

Two properties that commonly distinguish metals from non-metals are:

1. **Electrical conductivity:** Metals generally conduct electricity well because their delocalized electrons can move and carry charge. Most non-metals are poor conductors; **graphite is an exception.**
2. **Malleability and ductility:** Metals can generally be hammered into sheets and drawn into wires. Their layers of ions can shift while the attraction to delocalized electrons continues. Solid non-metals are generally brittle.

Q4. Oxidizing and reducing agents

Rule: The substance that is **reduced** is the **oxidizing agent**. The substance that is **oxidized** is the **reducing agent**.



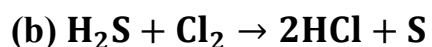
In WO_3 , each oxygen has oxidation state -2 . Thus, tungsten is $+6$, because $+6 + 3(-2) = 0$.

Element Oxidation-state change Process

W $+6$ in $\text{WO}_3 \rightarrow 0$ in W **Reduction**

H 0 in $\text{H}_2 \rightarrow +1$ in H_2O **Oxidation**

Answer: WO_3 is the oxidizing agent; H_2 is the reducing agent. Oxygen remains -2 .



Hydrogen is $+1$ in H_2S , so sulfur is -2 . Chlorine is 0 in elemental Cl_2 .

Element Oxidation-state change Process

S -2 in $\text{H}_2\text{S} \rightarrow 0$ in S **Oxidation**

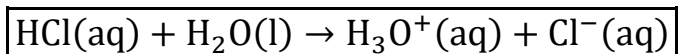
Cl 0 in $\text{Cl}_2 \rightarrow -1$ in HCl **Reduction**

Answer: Cl_2 is the oxidizing agent; H_2S is the reducing agent. Hydrogen remains $+1$.

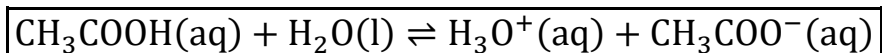
Q4 — OR: Strong acid versus weak acid

A **strong acid** ionizes **almost completely** in water. A **weak acid** ionizes **only partially**; most of its molecules remain un-ionized at any instant. “Strong” and “weak” describe **extent of ionization**, not how concentrated the acid solution is.

Hydrochloric acid is strong:



Ethanoic acid is weak:



The one-way arrow for HCl shows nearly complete ionization; the equilibrium arrow for CH₃COOH shows partial ionization. For the **same initial molar concentration**, HCl therefore produces more H₃O⁺ ions than ethanoic acid.

Q5. Exothermic, endothermic and enthalpy change

(a) Definitions

An **exothermic reaction releases heat** to its surroundings. Its enthalpy change is **negative**, $\Delta H < 0$. An example is combustion of methane.

An **endothermic reaction absorbs heat** from its surroundings. Its enthalpy change is **positive**, $\Delta H > 0$. An example is thermal decomposition of calcium carbonate.

(b) Calculate ΔH for $\text{H}_2 + \text{F}_2 \rightarrow 2\text{HF}$

Use:

$$\Delta H = \sum (\text{bond energies of bonds broken}) - \sum (\text{bond energies of bonds formed})$$

Broken: One H–H and one F–F bond:

$$436 + 155 = 591 \text{ kJ mol}^{-1}$$

Formed: Two H–F bonds:

$$2(567) = 1134 \text{ kJ mol}^{-1}$$

Therefore:

$$\Delta H = 591 - 1134 = -543 \text{ kJ mol}^{-1}$$

The negative value shows that formation of two H–F bonds releases more energy than is needed to break the reactant bonds. The reaction is **exothermic**.

Q5 — OR: Trends down Group I-A

a. Atomic radius increases. Each step down the group adds another occupied electron shell. The outer electron is therefore farther from the nucleus. Greater shielding by inner electrons also reduces the nucleus's effective pull on that outer electron. Thus:

$$\text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs in atomic radius}$$

b. Reactivity increases. Group I-A metals react by **losing their single outer electron** to form a +1 ion. Farther down the group, that electron is farther from the nucleus and more shielded, so it is **easier to remove**. Consequently, the metals generally react more readily:

$$\text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs in reactivity}$$

For example, potassium reacts more vigorously with water than sodium.